
Tracing LLM Reasoning Processes with Strategic Games: A Framework for Planning, Revision, and Resource-Constrained Decision Making

Xiaopeng Yuan^{*} Xingjian Zhang Ke Xu Yifan Xu Lijun Yu Jindong Wang

Yushun Dong

Haohan Wang[†]

Abstract

Large language models (LLMs) are increasingly applied to tasks that require complex reasoning. While most benchmarks focus on evaluating final reasoning outcomes, they overlook the internal processes that lead to those outcomes—such as how a model plans, revises, and makes decisions under constraints. We argue that evaluating these internal reasoning steps is essential for understanding model behavior and improving reliability in real-world applications. To make these processes observable and measurable, we propose using strategic games as a natural and effective environment. These games operate within closed, rule-based systems and provide interpretable states, limited resources, and automatic feedback.

Therefore, we propose a framework to evaluate LLMs along three core process dimensions: planning, revision, and resource-constrained decision making. To support this, we introduce a set of evaluation metrics that extend beyond traditional win rates, incorporating measures such as Over-correction Risk Rate, correction success rate, improvement slope, and over-budget ratio.

In a set of 4320 adversarial rounds across 12 state-of-the-art models, we find that ChatGPT-o3-mini, which demonstrates strong planning capabilities, achieves the highest composite process score (74.7% win rate, 78.6% correction success, and a +0.041 improvement slope). In contrast, Qwen-Plus, despite a high Overcorrection Risk Score of 81.6%, wins only 25.6% of its matches, primarily due to excessive resource use. We also observe a negative correlation between Over-correction Risk Rate and correction success rate (Pearson $r = -0.51$, $p = 0.093$), suggesting that more frequent corrections do not always improve outcomes. This pattern may reflect impulsive revision strategies, where premature edits reduce overall effectiveness, while more selective approaches lead to greater accuracy. We hope this work offers a new direction for LLM evaluation—focusing not just on what models decide, but on how they decide it.

1 Introduction

Large language models (LLMs) are now capable of solving increasingly complex reasoning tasks [14, 34]. As their performance on traditional benchmarks improves, it has become clear that measuring outcome accuracy alone is no longer sufficient. In many real-world scenarios, the quality of an LLM’s reasoning depends not only on the final answer, but also on the internal processes it uses to arrive there: how it plans, how it revises mistakes, and how it makes decisions under resource constraints.

^{*} First author. xyuan75@g.ucla.edu

[†] Corresponding author. haohanw@illinois.edu